SPECTRA: A Traceability Ontology for the 3GPP RAN Standardization Process

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Abstract. The 3rd Generation Partnership Project (3GPP) governs global mobile communication standards through a large document ecosystem spanning five Radio Access Network (RAN) Working Groups, yet, to the best of our knowledge, no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle and its multi-hop traceability process. We present **SPECTRA** (*SPECification TRAcing*), a reusable OWL ontology for the 3GPP RAN standardization process, organized around four concerns: contributions, resolutions, specifications, and cross-group communication. SPECTRA is grounded in 137 competency questions elicited from practicing standards engineers; representative CQs and their Cypher/SPARQL translations accompany the release so that the schema can be inspected and reused without access to the 3GPP source corpus. The ontology's quality is supported by a cumulative CQ validation protocol in which each of five incremental development phases extends the ontology for a new document type and the accumulated CQ suite is re-validated against the resulting schema state. Its applicability is demonstrated by cross-working-group instantiation across all five RAN Working Groups (RAN1–RAN5): the RAN1-designed schema required no additional classes, one additional object property, and two genuinely new data properties to cover RAN2–RAN5. The ontology currently serves as the semantic layer of a standards-engineering knowledge infrastructure at Samsung System LSI.

Keywords: Ontology engineering · Knowledge graph · 3GPP · Competency questions · Telecom standardization

1 Introduction

The 3rd Generation Partnership Project (3GPP) is the global standards body responsible for defining mobile communication technologies, from 4G LTE and 5G NR to the upcoming 6G standards. The standardization process is contribution-driven, with member companies submitting technical documents through 3GPP's Working Group (WG) structure. The RAN (Radio Access Network) Technical Specification Group alone encompasses five WGs (RAN1–RAN5), each responsible for a distinct layer of the radio interface — from physical layer procedures to conformance testing.

3GPP specifications are the normative reference that commercial 5G implementations and conformance tests must align with. Yet the corpus spans hundreds of normative specifications and evolves through a process that produces hundreds of thousands of Working Group documents. Acquiring even partial working knowledge typically demands graduate-level expertise and months of immersion, creating a barrier that compounds along several axes: *cross-WG* (a RAN1 decision propagates to RAN2/3/4 through Liaison Statements few engineers track end-to-end), *cross-organization* (each company has internal context only for its own contributions), and *industry–academia* (researchers see the published specifications but not the meeting-level decisions that shaped them). The traceability path TDoc → Resolution → CR → TS Section → Spec, which captures *why* a clause reads as it does and where it resides, currently exists as implicit institutional knowledge and is typically reconstructed by hand — error-prone and time-consuming.

To the best of our knowledge, no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle. Recent 3GPP-domain resources — TSpec-LLM [9] for retrieval-augmented access to specification text and *telecom-kg-rel19* [18] for spec-text concept extraction — model the *content* of specifications but are not designed to directly represent the standardization *process* that produced them — e.g., which TDocs led to a clause, which CRs were triggered by a Liaison Statement, or which Working Assumptions were promoted to Agreements.

In this paper, we present **SPECTRA** (*SPECification TRAcing*), a reusable OWL ontology for modeling the types, relationships, and lifecycle of 3GPP RAN standardization documents. The contributions are:

1. **The SPECTRA OWL ontology** — a reusable OWL resource for 3GPP RAN standardization documents and their lifecycle, comprising 26 classes and 134 properties (53 object properties, 81 data properties) organized around contributions, resolutions, specifications, and cross-group communication; the ontology reuses Dublin Core and FOAF and is licensed CC-BY 4.0 (release details in Section 7).
2. **A competency-question grounding** — SPECTRA's classes, properties, and axioms are derived from 137 competency questions elicited from practicing 3GPP standards engineers at Samsung System LSI; representative CQs with Cypher/SPARQL translations accompany the public release for inspection and reuse without access to the 3GPP source corpus, while the complete suite is retained as internal validation material.
3. **Cross-WG applicability and instantiation evidence** — the schema has been instantiated at realistic scale across all five RAN Working Groups (RAN1–RAN5), and applying the RAN1-designed schema to RAN2–RAN5 required *no additional classes, one additional object property, and two genuinely new data properties* (Table 7). The instantiated knowledge graphs and operational parsing pipeline are not redistributed because the underlying TDocs are subject to 3GPP copyright.

4. **A cumulative CQ validation protocol** as the quality-assurance mechanism behind the released ontology — whereas prior CQ-driven methodologies such as LOT [12] and SAMOD typically apply CQ validation within a single design iteration, we develop SPECTRA through five incremental phases (introducing 25, 34, 45, 15, and 18 CQs respectively, for a total of 137 distinct CQs); the accumulated CQ suite is re-validated against the new schema state at each phase transition. Three quality-assurance cases surfaced by this protocol were resolved before release, after which our internal validation records did not surface further regressions at subsequent phase transitions.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 provides background on 3GPP standardization. Section 4 describes the ontology design. Section 5 presents the CQ-driven cumulative validation methodology. Section 6 evaluates the ontology. Section 7 discusses availability and FAIRness. Section 8 concludes.

2 Related Work

2.1 Ontologies for Domain Knowledge Management

Ontologies have been widely adopted as the semantic backbone for domain-specific knowledge management systems. In cultural heritage, the Polifonia Ontology Network [2] provides 15 modular ontologies for musical heritage, developed through eXtreme Design with 361 competency questions from 10 pilots. In the energy domain, the Hydrogen Ontology (HOLY) [1] models the structure of hydrogen-based markets through CQ-driven design. In industry, Cao et al. [4] describe a knowledge graph for the quality management of Bosch products. For technical standards, Grangel-González and Vidal [6] use the Standards Ontology (STO) as the schema for an Industry 4.0 Knowledge Graph covering more than 200 industrial standards, but their focus is on *inter-standard* relationships rather than the *intra-standard* document structure central to our work. In telecommunications, NORIA-O [16] targets anomaly detection and incident management in ICT systems rather than the standardization process itself. The most directly comparable recent effort in the 3GPP domain is *telecom-kg-rel19* [18], a knowledge graph extracted from Release-19 specification *text* for retrieval-augmented LLM reasoning; SPECTRA differs in that it models the standardization *process and document lifecycle* (TDoc, Resolution, CR, LS, TR, Section, etc.) rather than concept structures embedded in spec text. To the best of our knowledge, no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle and traceability process.

2.2 Ontology Engineering Methodologies

The Ontology 101 guide [10] established a widely-adopted methodology with competency questions (CQs) as the primary requirements mechanism [7]. LOT [12] extends this for industrial settings with FAIR compliance, the NeOn

methodology [15] provides scenario-based pathways for building ontology networks (including reuse of existing ontological and non-ontological resources), and SAMOD [11] draws on Test-Driven Development principles. While these methodologies support iterative refinement and, in NeOn's case, modular network development, they do not explicitly frame CQ validation as a *cumulative* protocol across multiple development phases — extending an ontology for a new sub-domain while re-executing the full CQ suite accumulated from prior phases, so that no earlier competency is silently regressed. SPECTRA's five-phase cumulative CQ validation (Section 5) uses this protocol as a quality-assurance mechanism for the released resource rather than a replacement for existing methodologies.

In terms of CQ scale, the Polifonia Ontology Network [2] releases a 361-CQ dataset (*PolifoniaCQ*) collected across the network's modules, and Wisniewski et al. [17] analyzed 234 CQs from 5 ontologies and translated them into 131 SPARQL-OWL queries. SPECTRA contributes 137 CQs applied cumulatively to a single cohesive ontology covering five document domains.

2.3 LLM-assisted Ontology Engineering

Large language models have recently been explored as assistants for ontology engineering: Saeedizade and Blomqvist [14] report that GPT-4 reaches "novice human modeler" quality for CQ-driven OWL generation on small general-purpose ontologies, and recent systematic reviews discuss a growing inventory of LLM-assisted ontology engineering tasks. The present paper focuses on the SPECTRA resource itself; systematic comparison against LLM-generated counterparts for specialized industrial domains is a complementary direction that we leave to future work.

3 3GPP Standardization Background

3.1 Document Types and Lifecycle

The 3GPP standardization process produces several distinct document types. **Technical Documents (TDocs)** are the primary unit of contribution, carrying structured metadata including a unique identifier (e.g., R1-2400001), submitting company, target agenda item, and purpose. **Resolutions** record meeting decisions as Agreements, Conclusions, or Working Assumptions. **Change Requests (CRs)** formally propose modifications to specifications. **Technical Specifications (TSs)** are normative documents (e.g., TS 38.211), organized hierarchically into sections. **Technical Reports (TRs)** document feasibility studies. **Liaison Statements (LSs)** are official cross-WG communications.

3.2 Working Group Structure

The RAN TSG is divided into five Working Groups:

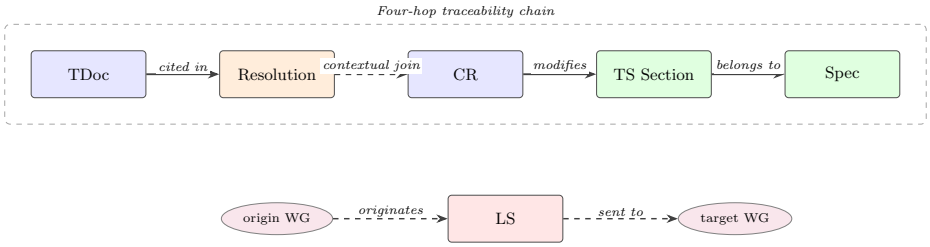

Fig. 1. Document lifecycle in 3GPP RAN. The top row is a reader-facing traceability path from a TDoc contribution to the affected specification: `references` is formally defined Resolution → Tdoc and is drawn in the reading direction, `modifiesSection` links CRs to Sections, and `belongsToSpec` links Sections to Specs. The dashed Resolution–CR step denotes an indirect join through meeting/agenda and CR-pack context rather than a single asserted ontology property. The bottom row summarizes LS-based cross-WG communication via `originatedFrom` and `sentTo`.

WG	Responsibility	Example Specs
RAN1	Physical layer (PHY)	TS 38.211–214
RAN2	Layer 2/3 protocols	TS 38.331, 38.321
RAN3	Architecture and interfaces	TS 38.401, 38.423
RAN4	Radio performance and RF	TS 38.101, 38.104
RAN5	Conformance testing	TS 38.521, 38.533

Changes in one WG frequently trigger Liaison Statements to other WGs, creating cross-group dependencies. RAN WG meetings convene roughly every two months, with each meeting typically generating thousands of TDocs.

3.3 Concepts an Ontology Must Capture

Three features of the corpus shape the ontology: **scale** (any single WG already accumulates on the order of 10^5 TDocs over its meeting history, so the schema must stay compact while admitting large-scale instantiation), **document lineage** (the TDoc→Resolution→CR→Section→Spec path demands a first-class entity and directed links for each document type), and **cross-group dependencies** (Liaison Statements must be modeled as directed artefacts between Working Groups rather than as free-form text). SPECTRA's class and property layout (Section 4) is organized around these three features.

4 SPECTRA Ontology Design

4.1 Competency Question Analysis

The design of SPECTRA is driven by 137 competency questions collected across five development phases, in collaboration with domain experts — standardiza-

tion engineers at Samsung System LSI who implement 3GPP specifications in commercial chipsets.

Table 1. CQ distribution across ontology development phases.

Phase	Domain	CQs	Example CQ
1	TDoc metadata	25	"What TDocs did Company X submit?"
2	Meeting resolutions	34	"What agreements on Agenda Item Z?"
3	TS structure	45	"Which CRs modified section S?"
4	CR documents	15	"What is the reason for change of CR C?"
5	Technical reports	18	"Which TRs have impacted TS 38.211?"
	Total	**137**	

4.2 Core Schema

Based on the CQ analysis, we designed SPECTRA following the Ontology 101 methodology [10]. The ontology is encoded in OWL 2 [8] and comprises 26 classes, 53 object properties, and 81 data properties, organized around four conceptual groups (Figure 2).

Document Classes. The central class is `Tdoc`, with four subclasses: `CR` (Change Request), `LS` (Liaison Statement), `Summary`, and `SessionNotes`. CRs have `modifies` (linking to a Spec), while LSs have `sentTo` and `originatedFrom` (linking to WorkingGroups).

Resolution Classes. `Resolution` has three subclasses: `Agreement`, `Conclusion`, and `WorkingAssumption`. A WorkingAssumption may be `promotedTo` an Agreement.

Specification Classes. `Spec` represents Technical Specifications; `Section` represents individual sections, forming a tree via `hasSubSection` / `parentSection` (both irreflexive and asymmetric). `TechnicalReport` links to an influenced Spec through a reified `TRImpact` class so a single TR can affect multiple Specs without losing per-impact metadata.

Organizational Classes. `Meeting`, `Company` ($\sqsubseteq$ `foaf:Organization`), `Contact` ($\sqsubseteq$ `foaf:Person`), `WorkItem`, `AgendaItem`, `Release`, and `WorkingGroup`.

Key Design Decisions. **D1:** Document types are modeled as OWL subclasses (not property values), enabling type-specific properties. **D2:** The base class was renamed from `Decision` to `Resolution` during integrity verification to align with 3GPP terminology. **D3:** 23 object properties carry `owl:FunctionalProperty` to declare an *at-most-one* contract (e.g., a TDoc is presented at at most one meeting); the loader uses this declaration to detect violations and also checks minimum-cardinality, which is not enforced at the OWL level. **D4:** The traceability spine (Figure 2) is realized by three direct object properties (`references`, `modifiesSection`, `belongsToSpec`) and a multi-hop join (Resolution $\rightarrow$ CR via

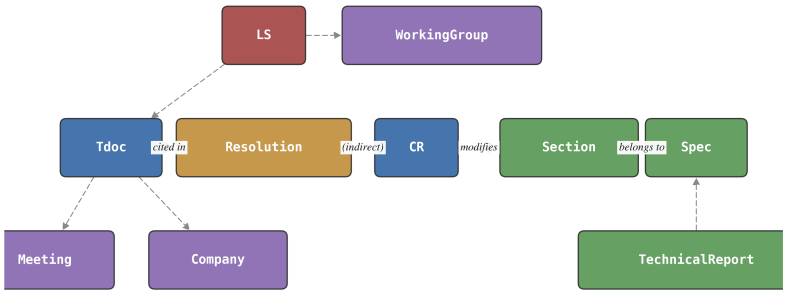

Fig. 2. SPECTRA core schema overview (reader-facing labels in the figure; formal object-property names below). **Central row**: the traceability spine `Tdoc` → `Resolution` → `CR` → `Section` → `Spec`. "*cited in*" = `references` (defined Resolution → Tdoc, drawn reversed); "*modifies*" = `modifiesSection`; "*belongs to*" = `belongsToSpec`. The Resolution → CR arrow is drawn lightly because the released ontology does not declare a direct property between these classes; the transition is reconstructed by joining `resolutionBelongsTo` (Resolution → AgendaItem) with `belongsToCRPack` (CR → CRPack) at the same meeting. **Dashed grey arrows** (unlabeled for clarity) carry secondary relations: `LS` → `Tdoc` is `rdfs:subClassOf`; `LS` → `WorkingGroup` is `sentTo`; `Tdoc` → `Meeting` is `presentedAt`; `Tdoc` → `Company` is `submittedBy`; `TechnicalReport` → `Spec` is `impactsSpec` (mediated by the reified `TRImpact` class). The released ontology declares 26 classes and 134 properties; remaining classes (other `Tdoc`/`Resolution` subclasses, `Contact`, `WorkItem`, `AgendaItem`, `Release`) and FOAF reuse are described in Section 4.

`resolutionBelongsTo`/`belongsToCRPack`); we avoid chain-level OPs so each declared property keeps a single, verifiable semantic, and the spine remains a reading aid rather than a single asserted relation.

4.3 Reuse of Existing Vocabularies

SPECTRA reuses established Semantic Web vocabularies where they naturally apply — specifically, Dublin Core terms [5] for metadata and FOAF [3] for organizational entities — and introduces domain-specific classes only where no suitable external term exists. Table 2 summarizes the reuse.

Domain-specific classes (`Tdoc`, `Resolution`, `Spec`, `CR`, `LS`, `TechnicalReport`, ...) are introduced because no external Semantic Web vocabulary models 3GPP document types at the granularity required by the CQs.

4.4 Constraints and Axioms

SPECTRA includes 23 `owl:FunctionalProperty` (at-most-one cardinality); 2 `owl:InverseFunctionalProperty` (`hasCR`, `hasTRImpact`); 15 inverse property pairs; 6 `owl:IrreflexiveProperty` covering the section hierarchy, intra-section references, and TR cross-references; and 2 `owl:AsymmetricProperty` on the

Table 2. External vocabulary reuse in SPECTRA.

Vocabulary	Terms reused	Role in SPECTRA
Dublin Core (`dc:`)	`title`, `description`, `creator`, `date`	Ontology and instance metadata
FOAF (`foaf:`)	`Person`, `Organization`	Parent classes for `Contact` and `Company`
OWL 2 (`owl:`)	`Class`, `FunctionalProperty`, `InverseFunctionalProperty`, `inverseOf`, `IrreflexiveProperty`, `AsymmetricProperty`	Formal axioms and property characteristics
RDF Schema (`rdfs:`)	`subClassOf`, `label`, `comment`	Class hierarchy and documentation

section hierarchy. These exclude self-loops and 2-cycles in the affected relations; longer cycles are prevented by loader-level checks.

5 CQ-Driven Cumulative Validation

The cumulative CQ protocol introduced in this section is the quality-assurance mechanism behind the released ontology: each new phase must leave every previously validated competency intact, so that schema refinements are caught at the point of introduction rather than at release.

5.1 Five-Phase Progressive Construction

SPECTRA was developed through five incremental phases, each extending the ontology to cover a new document type while re-executing the full CQ suite accumulated from prior phases. The ontology grew from 11 classes (Phase 1–2, TDoc metadata) to 26 classes (Phase 5, Technical Reports), with the final v6.1.1 release incorporating refactoring identified through cross-phase integrity verification.

5.2 Quality-Assurance Cases

Rerunning the accumulated CQ suite at each phase transition surfaced concrete schema issues. Three representative cases (each traceable via git history): **Case 1** (Phase-3 transition) — a relationship-naming inconsistency (`MADEAT` vs `MADE_AT`); three relationship and three property names were harmonized. **Case 2** (Phase-3 audit) — a meeting-ID inconsistency affecting referential integrity; three targeted patches resolved it without schema change. **Case 3** (Phase-3→Phase-4) — three schema refinements: (i) `releaseId` removed as redundant with `releaseName`, (ii) `hasCR` received an `owl:InverseFunctionalProperty` declaration, (iii) the base class `Decision` was renamed `Resolution` to align with 3GPP terminology. The cumulative protocol thus functions as a gating mechanism — gaps a single-phase validation might miss are caught by re-execution against earlier CQ sets.

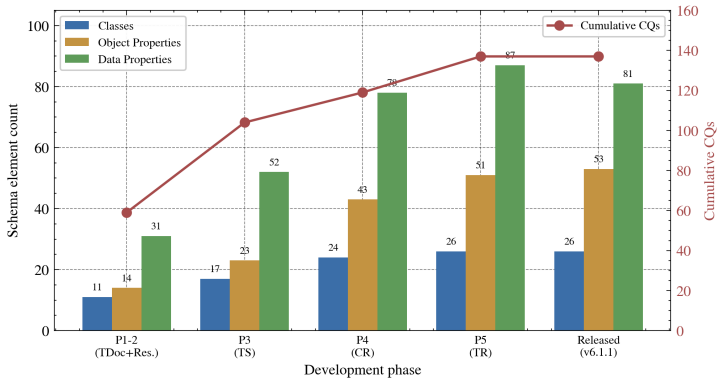

Fig. 3. Schema growth across the five-phase development of SPECTRA, measured from git snapshots at each phase-closing commit. Cumulative CQs (right axis) reach 137 by Phase 5. The data-property drop between Phase 5 and the released version reflects refactoring described in Section 5.

5.3 Cumulative Regression Verification

At each phase transition, all CQs from Phase 1 through Phase N are re-validated against the schema state introduced by Phase N, executed over an internally instantiated knowledge graph that conforms to the schema. The 137 distinct CQs accumulated by the end of Phase 5 (25 + 34 + 45 + 15 + 18) all return correct, non-empty results against this internal KG (Section 6, Table 3). After the quality-assurance cases above were resolved, our internal validation records did not surface further regressions at subsequent phase transitions; this is an internal observation, since the instantiated KG is not part of the public release.

6 Evaluation

6.1 CQ Coverage

All 137 CQs return correct, non-empty results when executed against the internally instantiated knowledge graph that conforms to the released schema. This 100% pass rate reflects the outcome of multiple refinement cycles (Section 5).

6.2 Structural Quality

We report structural metrics of the released ontology (Table 4) and the findings of an automated pitfall scan (Table 5).

We evaluated SPECTRA with OOPS! [13] via its REST API. The scan reports **no critical pitfalls**, two *Important*, and two *Minor* (Table 5). Both *Important* findings are expected consequences of design decisions: **P34** flags `foaf:Person/foaf:Organization` (we reuse FOAF without redeclaring, the

Table 3. CQ validation results across ontology phases.

Phase	Domain	CQs	PASS	FAIL	Rate
1	TDoc metadata	25	25	0	100%
2	Meeting resolutions	34	34	0	100%
3	TS structure	45	45	0	100%
4	CR documents	15	15	0	100%
5	Technical reports	18	18	0	100%
	Total	**137**	**137**	**0**	**100%**

Table 4. Ontology structural metrics.

Metric	Value
Classes	26
Object Properties	53
Data Properties	81
Functional Properties	23
Inverse Functional Properties	2
Inverse Property Pairs	15
Irreflexive Properties	6
Asymmetric Properties	2
Reused vocabulary terms	6 (Dublin Core, FOAF)

recommended pattern), and **P10** flags missing `owl:disjointWith` between document-type classes (a conservative choice because the released CQs do not rely on disjointness, and adding it without instance-level verification risks reasoner inconsistencies). The two *Minor* pitfalls — P08 (annotations on reused FOAF classes; handled by the source vocabulary) and P13 (inverses declared only for bidirectionally queried pairs) — reflect the same reuse-and-parsimony posture.

Table 5. OOPS! results for the released SPECTRA ontology.

Code	Name	Severity	# affected
P08	Missing annotations (on reused FOAF classes)	Minor	2
P10	Missing disjointness (between document-type classes)	Important	1
P13	Inverse relationships not explicitly declared	Minor	23
P34	Untyped class (reused FOAF classes)	Important	2

The release also ships a PyLODE-rendered HTML of the ontology and a SHACL shapes file (`spectra-core.shacl.ttl`) covering core cardinality/range constraints, validated against the bundled synthetic instantiation via pySHACL.

6.3 Instantiation at Scale

To provide internal evidence that the schema tolerates realistic document volumes, we instantiated SPECTRA against RAN1 data. The graph contains 123,677 Technical Documents from 60 meetings, together with all associated Resolutions, Change Requests, Liaison Statements, Specifications, and Technical Reports (Table 6). Referential integrity for the `submittedBy` relation, the most heavily relied on by the competency questions, is 99.36% (1,000 missing links out of 155,525 `Tdoc`–`Company` associations); the residual gap is traceable to missing source metadata in the original 3GPP TDoc list rather than to schema inadequacy. This instantiation is maintained as internal validation evidence; per Section 7 it is not part of the public release because the underlying TDocs are subject to 3GPP copyright.

Table 6. Per-class instance counts in the RAN1 instantiation of SPECTRA (internal validation evidence).

Class	**#**	**Class**	**#**	**Class**	**#**
Tdoc	123,677	Resolution	24,710	Section	1,003
CR	10,715	Agreement	21,058	TSTable	680
LS	6,343	Conclusion	2,703	TSFigure	13
Summary	3,468	WorkingAssump.	949	CRPack	501
SessionNotes	678	Meeting	60	TechnicalReport	26
Spec	83	Company	222	TRImpact	29
Release	13	Contact	1,000	Figure	120
WorkingGroup	116	WorkItem	423	Table	1,441
AgendaItem	6,439			Chart	1

6.4 Cross-WG Applicability and Reuse

A schema designed for one WG can fail silently when applied to a sibling WG with different document conventions. We applied SPECTRA — originally designed from RAN1 CQs — to RAN2–RAN5 without redesign and measured a stricter quantity than label overlap: how many *additional* schema elements each WG needed beyond what the RAN1-designed ontology already defined. Table 7 reports the schema-level diff computed from each WG's instantiated KG.

The 26 classes cover every document-type distinction the per-WG loaders produced (**zero** additional classes); only one additional object property (`REFERENCES_SPEC`) appears across all four non-RAN1 WGs. The 22 apparent data-property extensions decompose into 20 loader-local attributes (17 underscore-prefixed mirror slots, e.g., `_submittedBy`; 3 scaffolding properties: `id`, `release`, `for_` — none of which represent ontology-level commitments) and two genuinely new data properties (`discussionText`, `summaryId`). The total additional schema surface across four sibling WGs is therefore one OP and two DPs on top of the 160 elements already declared — evidence that the RAN1 schema is stable under cross-WG application within the RAN TSG rather than a universal transferability claim.

Table 7. Schema diff of RAN2–RAN5 instantiations against the RAN1-designed SPECTRA schema. "WG-specific extensions" counts schema elements present in a non-RAN1 WG graph but not declared in the released ontology (union over RAN2–RAN5).

Schema element	In released SPECTRA	WG-specific extensions	Extension rate
Classes	26	0	0.0%
Object Properties	53	1	1.9%
Data Properties	81	2 †	2.5%

† excluding 20 loader-local properties (17 underscore-prefixed mirror slots and 3 internal scaffolding properties).

Conversely, 6 of the 26 RAN1 classes are absent from at least one sibling WG's instantiation (e.g., RAN5 is scoped to conformance testing and populates 20 of the 26 classes). These absences are expected and not counted as failures: they indicate that a non-RAN1 WG does not need to populate the full RAN1 ontology to benefit from reuse.

6.5 Representative Use Scenarios

To illustrate what the ontology is for, we list a curated set of competency questions that SPECTRA supports against an instantiated knowledge graph; each question is paired with the schema fragment it exercises and the role it plays in the standardization workflow.

- **Multi-hop traceability.** "Which TDoc motivated the current text of TS 38.214 §5.1.3?" — starts at the Section, traverses CRs that modify it (reverse traversal over `modifiesSection`), reaches the Resolutions linked through the indirect join of Figure 1, and finally retrieves the TDocs cited by those Resolutions (reverse traversal over `references`).
- **Cross-WG impact.** "Which RAN2 contributions were triggered by a RAN1 Liaison Statement in meeting M?" — uses `LS.originatedFrom` / `sentTo` between `WorkingGroup` nodes.
- **Release-scoped queries.** "What are all CRs against Release 18 submitted by Company X?" — combines `CR.targetRelease` (CR is a `Tdoc` subclass; `targetRelease` has domain `Tdoc` ⊔ `TechnicalReport`), `CR.modifies`, and `Tdoc.submittedBy`.
- **Resolution lifecycle.** "Which Working Assumptions were promoted to Agreements over the last three meetings?" — follows `WorkingAssumption.promotedTo` through `Resolution` subtypes.

The public release ships Cypher and SPARQL translations of these representative queries so that reusers can exercise the schema on their own instance data. The full 137-CQ validation suite is retained as internal validation material (Section 5).

7 Availability, Sustainability, and FAIRness

We distinguish explicitly between the *publicly released resource* and the *internal validation evidence* on which it rests, because parts of the latter cannot be redistributed due to 3GPP copyright on source documents and confidentiality constraints on the operational pipeline.

Publicly released (CC-BY 4.0).

- **SPECTRA ontology** (OWL 2 Turtle), persistent IRI `https://w3id.org/spectra`; **PyLODE HTML documentation** and a **SHACL shapes file** for core cardinality/range constraints.
- **README** (scope, version, license) and machine-readable metadata (`CITATION.cff`, `codemeta.json`); a schema diagram.
- **Representative competency questions** (curated subset of the 137-CQ suite) with **Cypher and SPARQL translations** and a **synthetic instantiation snippet**.

Not publicly released.

- The complete 137-CQ validation suite and the full set of executable Cypher queries used for cumulative regression testing (retained as internal validation material; representative subsets are released as noted above).
- The instantiated 3GPP RAN knowledge graphs (per-WG instantiations at realistic scale; not redistributable because the underlying TDocs are subject to 3GPP copyright; the original TDocs remain publicly accessible via the 3GPP portal at `https://www.3gpp.org`).
- The operational parsing and knowledge-graph population pipeline (internal project code; not released).

FAIR alignment. **Findable:** a persistent Zenodo DOI will be minted from a tagged GitHub release upon camera-ready publication; the GitHub repository hosting the release package is already prepared. **Accessible:** the OWL file and documentation are designed to be downloadable without authentication once the release is published. **Interoperable:** encoded in OWL 2 (Turtle), loadable by standard tooling (Protégé, Apache Jena, RDFLib); reuses Dublin Core and FOAF (Section 4). **Reusable:** licensed under CC-BY 4.0, with version information, a README, and representative query examples.

Sustainability. The ontology currently serves as the semantic layer of a standards-engineering knowledge infrastructure at Samsung System LSI, which motivates ongoing maintenance: incremental updates for each new 3GPP meeting cycle are produced internally, and the public ontology release is versioned accordingly. Changes to the public ontology are validated against the cumulative CQ protocol before release.

Resource Availability Statement. The release package (SPECTRA ontology in OWL/Turtle, README, schema diagram, representative CQ examples, and representative Cypher/SPARQL translations) will be published on Zenodo with a persistent DOI minted from a tagged GitHub release upon camera-ready publication; the public DOI and release URL will replace this statement at that time. The complete 137-CQ validation suite, the full executable query set, the instantiated knowledge graphs, and the operational parsing pipeline are not part of the public release. The underlying 3GPP Technical Documents are subject to 3GPP copyright and remain publicly accessible via the 3GPP portal (`https://www.3gpp.org`).

8 Conclusion

We presented **SPECTRA** as a reusable OWL ontology resource for 3GPP RAN standardization, released under CC-BY 4.0 (Zenodo DOI upon publication) and accompanied by a representative subset of the competency questions and example Cypher/SPARQL translations used in its validation. The resource was developed through a five-phase cumulative CQ validation protocol covering 137 competency questions, with three quality-assurance cases resolved before release; an internally instantiated KG conforming to the schema answers all 137 CQs, and applicability is demonstrated by cross-WG instantiation across RAN1–RAN5 (with non-redistribution of the instantiated KG and pipeline as detailed in Section 7).

Limitations and future work. The scope is RAN TSG; extending to TSG SA/CT would add classes and relationships, and the publicly released queries are representative rather than exhaustive. We plan to extend to TSG SA/CT, investigate LLM-assisted CQ generation, and examine other standards bodies (IETF, IEEE).

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References

1. Ascencion Arevalo, K.M., Neunsinger, C., Zimmermann, R., Blum, R., Weakly, K.: HOLY: An ontology covering the hydrogen market. In: The Semantic Web – ISWC 2023, Resources Track. Lecture Notes in Computer Science, Springer (2023), ontology release: `https://purl.org/holy`
2. de Berardinis, J., Carriero, V.A., Jain, N., Lazzari, N., Meroño-Peñuela, A., Poltronieri, A., Presutti, V.: The Polifonia Ontology Network: Building a semantic backbone for musical heritage. In: The Semantic Web – ISWC 2023. pp. 302–322. Springer (2023)
3. Brickley, D., Miller, L.: FOAF Vocabulary Specification 0.99. Namespace Document, 14 January 2014 (2014), `http://xmlns.com/foaf/spec/`

4. Cao, Q., Grangel-González, I., Du, L.: A knowledge graph-based approach for the quality management of Bosch products. In: ISWC 2023 Posters, Demos, and Industry Tracks. CEUR Workshop Proceedings, vol. 3632 (2023), https://ceur-ws.org/Vol-3632/ISWC2023_paper_507.pdf
5. DCMI Usage Board: DCMI Metadata Terms. DCMI Recommendation, 20 January 2020 (2020), https://www.dublincore.org/specifications/dublin-core/dcmi-terms/
6. Grangel-González, I., Vidal, M.E.: Analyzing a knowledge graph of industry 4.0 standards (2021), manuscript; ontology and accompanying knowledge graph for Industry 4.0 standards
7. Gruninger, M., Fox, M.S.: The role of competency questions in enterprise engineering. In: Benchmarking—Theory and Practice. pp. 83–95. Springer (1995)
8. Hitzler, P., Krötzsch, M., Parsia, B., Patel-Schneider, P.F., Rudolph, S.: OWL 2 Web Ontology Language Primer (Second Edition). Tech. rep., W3C Recommendation, 11 December 2012 (2012), https://www.w3.org/TR/owl2-primer/
9. Nikbakht, R., Benzaghta, M., Geraci, G.: TSpec-LLM: An open-source dataset for LLM understanding of 3GPP specifications. In: IEEE Globecom Workshops (GC Wkshps) (2024)
10. Noy, N.F., McGuinness, D.L.: Ontology development 101: A guide to creating your first ontology. Tech. Rep. KSL-01-05, Stanford Knowledge Systems Laboratory, Stanford University (2001)
11. Peroni, S.: A simplified agile methodology for ontology development. In: OWL: Experiences and Directions – Reasoner Evaluation (OWLED 2016). Springer (2016)
12. Poveda-Villalón, M., Fernández-Izquierdo, A., Fernández-López, M., García-Castro, R.: LOT: An industrial oriented ontology engineering framework. Engineering Applications of Artificial Intelligence **111**, 104755 (2022)
13. Poveda-Villalón, M., Gómez-Pérez, A., Suárez-Figueroa, M.C.: OOPS! (OntOlogy Pitfall Scanner!): An on-line tool for ontology evaluation. International Journal on Semantic Web and Information Systems (IJSWIS) **10**(2), 7–34 (2014). https://doi.org/10.4018/ijswis.2014040102
14. Saeedizade, M.J., Blomqvist, E.: Navigating ontology development with large language models. In: The Semantic Web – ESWC 2024. Springer (2024)
15. Suárez-Figueroa, M.C., Gómez-Pérez, A., Fernández-López, M.: The NeOn Methodology for Ontology Engineering. In: Ontology Engineering in a Networked World, pp. 9–34. Springer (2012)
16. Tailhardat, L., Chabot, Y., Troncy, R.: NORIA-O: An ontology for anomaly detection and incident management in ICT systems. In: The Semantic Web – ESWC 2024. pp. 21–39. Springer (2024). https://doi.org/10.1007/978-3-031-60635-9_2
17. Wisniewski, D., Potoniec, J., Lawrynowicz, A., Keet, C.M.: Analysis of ontology competency questions and their formalizations in SPARQL-OWL. Journal of Web Semantics **59**, 100534 (2019)
18. Yang, Y., Bariah, L., Lu, Y., Debbah, M.: telecom-kg-rel19: A telecom knowledge graph from 3GPP release 19 specifications. Hugging Face Datasets, GSMA / Khalifa University Research Institute for Digital Future (2026), https://huggingface.co/datasets/GSMA/telecom-kg-rel19